

Gradient Descent and Logistic Regression

Lab 3

1. Create a dataframe for the following data:

X	1	2	3	4	5
Y	2	5	7	8	9

- a) Use alpha/learning rate = 0.01, and find the optimal value of slope and intercept for the above continuous data
 - b) Find the optimized equation
 - c) Determine MSE and R2-score for the above data.
2. Create a dataframe for the following data:

Study Hours	15	29	28	31	35
Result	Fail	Fail	Pass	Pass	Pass

- a) Create a logistic regression model for the above data.
 - b) Find its accuracy and classification metrics
3. Perform logistic regression on a Kaggle dataset. Search for any binary class data and try to perform logistic regression and find the classification metrics.